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Theories to Advance Dark Patterns Research

Thomas (Tomek) Schilling 

Department of Psychology, Trier University, Trier, Germany

ABSTRACT

In this article, I review general consumer theories related to dark patterns research, by highlighting their relevance to understanding how dark patterns affect consumer choices on digital platforms. I critique the inadequacy of current theoretical references in dark patterns research, which often relies on heuristics and biases or the dual-process model, and I argue that these concepts, referred to as theory, do not explain the mechanisms behind consumer responses to dark patterns. To advance theory-based dark patterns research, I present a broad overview of diverse consumer theories, such as the Theory of Planned Behavior, the Meta-Theoretic Model of Motivation and Personality, and the Stimulus-Organism-Response Theory, among others, to illustrate how these frameworks can inform empirical hypothesis testing and advance the understanding of consumer vulnerabilities. By synthesizing various theoretical perspectives, I provide a foundation for refining dark patterns taxonomies and informing the design of empirical dark patterns research. In conclusion, I emphasize the necessity for future research to develop and review specific theories that apply directly to different types of dark patterns. A nuanced theoretical approach will be necessary to better understand and mitigate the manipulative effects of dark patterns in consumer behavior.

KEYWORDS

Deceptive designs;
hypotheses development;
theory

1. Introduction

Dark patterns research to date has developed rapidly but remains theoretically under-integrated (Chang et al., 2024; Kitkowska, 2023). This is not an isolated issue, as several review articles highlighted a dearth of theory or mechanistical explanations of related research such as on persuasive technologies (Adams et al., 2015) and choice architecture research (Szasz et al., 2018). As these review articles laid out, many studies were performed without explicit theoretical grounding, and when theory was cited, references often included “nudge theory,” “heuristics and biases,” or “dual-process theory.” While these frameworks offer valuable psychological insights, they do not meet the definitional criteria of theory as used in this article: that is, a theory provides a structured set of propositions enabling causal explanation, behavioral prediction, and the derivation of empirically testable hypotheses (Burton, 2001, 2005; Gray, 2017; Sutton & Staw, 1995; Whetten, 1989; Zeithaml et al., 2020). A theory thus should allow one to make mechanistic causal claims between various measurable factors (variables, e.g., price and consumer demand) within a specified realm of activities (e.g., consumer decisions online); to make predictions of future behaviors or actions (e.g., “If we increase price today, sales will be lower from now on”) and to form empirically testable hypotheses (e.g., “A price increase leads to reduced sales”).

Throughout this article, I reserve the term dark patterns for interface designs that intentionally steer users toward choices misaligned with their best interests through deception, coercion, asymmetric information, and sometimes through mere friction or annoyance (Gray et al., 2024; Mathur et al., 2019). It is important to note that persuasion itself is neither inherently unethical nor problematic. Much influence in digital environments—such as reminders, recommendations, defaults, or simplifications—is

CONTACT Thomas (Tomek) Schilling  thomas.schilling@uni-trier.de  Department of Psychology, Trier University, Trier, Germany

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lawful, commonplace, and often welfare-enhancing for users. Behavioral insights are routinely used to reduce friction, support goal attainment, and improve decision outcomes. Dark patterns, however, tend to do the opposite. The present article does not treat all forms of steering, nudging, or UX friction as suspect. Rather, the concern with dark patterns arises when design practices systematically exploit psychological mechanisms in ways that are opaque and misaligned with users' interests, particularly when they undermine perceived control, bypass reflective choice, or disproportionately affect vulnerable users. Distinguishing benign influence from manipulation therefore requires attention not only to interface design features, but to the underlying mechanisms through which influence operates.

This article focuses on dark patterns in persuasive and commercial settings—particularly e-commerce, games, and app interfaces designed to shape consumer decisions. I do not claim universal applicability of consumer theories to all dark pattern contexts. Other domains—such as privacy, consent, or civic participation—may require different epistemological frameworks, including socio-technical systems or interaction criticism, which involve legal, ethical, experiential, and temporal aspects (Gray et al., 2021). My goal is to demonstrate how consumer theories can strengthen mechanistic, testable explanations in persuasion-oriented dark patterns research to support our understanding of their influence on consumers. I focus on consumer decision theories because most dark patterns in practice are employed to influence consumers' decisions (Di Geronimo et al., 2020; Mathur et al., 2019; Zagal et al., 2013).

Conceptual models like heuristics, dual-process theory, and choice architecture have undoubtedly shaped dark patterns research by offering shared language, highlighting cognitive vulnerabilities, and guiding empirical work. The field has made important contributions by documenting design tactics, developing taxonomies, and informing regulatory debates (Gray et al., 2024; Mathur et al., 2019). My goal is to build on these contributions—not to discredit them—by introducing a complementary, theory-based perspective focused on predictive and mechanistic explanation. Unlike conceptual models, theories in this sense allow researchers to formulate falsifiable hypotheses about which types of dark patterns influence which outcomes, through which psychological mechanisms, and in which people.

So, why do we need to apply decision theories? As has been highlighted previously, “it is unclear which types of deceptive design practices need to be better understood, which types of patterns produce the most harm, and what kinds of studied regulatory and legal professionals need to effectively identify, characterize, and sanction the use of dark patterns” (Gray et al., 2023, p. 188). Consumer decision theories can help to achieve all the points highlighted in this quote. Choice architecture, for example, is best understood as a conceptual lens or design philosophy rather than a predictive theory: it captures the idea that every artificially designed aspect of people's environments affects their choices and those who design the environments in which people act are “choice architects.” But it does not help to understand which design feature (dark pattern) might lead to what type of behavior or outcome (more or less harm). The dual-process model suggests that we have two information processing systems in our brains. This model does not predict which system is activated when or what factors contribute to the (de)activation of one system or the other, and the model also does not make any predictions regarding specific behaviors in specific contexts. Likewise, the heuristics and biases literature constitutes a broad taxonomy of mental shortcuts and associated errors—and there are too many of them within the heuristics and biases realm to mention them here, so I refer the interested reader to two Wikipedia articles¹—rather than a unified theory that yields predictions about behavior in specific choice environments.

To situate this article within the existing empirical literature and illustrate how theory is currently used in empirical dark patterns research, I reviewed 23 experimental studies on dark patterns (see online supplementary material).² This review revealed that only four studies explicitly referred to predictive theories that could yield testable hypotheses. While some researchers referenced psychological constructs such as trust, emotions, or perceived autonomy, they did so inconsistently—no two studies used the same measurement scales, and many relied on newly developed, unvalidated items. This inconsistency makes it difficult to compare findings or build cumulative knowledge. Researchers argued in their articles that dark patterns affect internal psychological states or exploit psychological mechanisms, which demonstrates that scholars are seeking to measure these effects—just barely grounded in formal theory. The present article seeks to build on that work by introducing relevant consumer behavior theories that could help unify findings and provide a stronger basis for variable selection, prediction, and interpretation in future studies.

Understanding the psychological mechanisms through which dark patterns influence consumer choices is critical—not only for scholars, but also for designers, marketers, and regulators. The consumer theories discussed here provide this critical explanatory depth to help generate empirically testable hypotheses, strengthen theoretical foundations, and inform more refined taxonomies. A deeper grasp of these mechanisms can help (1) platform designers to make more deliberate and ethically responsible design decisions and (2) regulators and researchers to identify the most harmful dark patterns and enable them to distinguish between benign nudges and manipulative practices that exploit human vulnerabilities. Theory, in the mechanistic sense described earlier, provides a structured framework for such understanding by offering explanatory pathways and testable predictions. It is my hope that this article brings relevant decision-theories to the attention of dark patterns scholars so that they become a central part of empirical studies on dark patterns. While much of the human-computer interaction (HCI) literature on dark patterns has advanced rich conceptual frameworks—drawing from methodologies such as interaction criticism (Gray et al., 2021) and emphasizing the temporality of user experience within socio-technical systems (Gray et al., 2025)—these approaches often differ in epistemological focus from those in consumer psychology, which tend to emphasize formal theory development and mechanistic modeling. Research in social psychology, marketing, and consumer psychology has a long tradition of engaging with formal theories that explain the mechanisms through which interface designs—including dark patterns, advertising, and other aspects of user experiences—shape consumer behavior.³ Because many dark patterns closely resemble persuasive marketing tactics that, for example, exploit cognitive biases (Dowling et al., 2020), dark patterns research may benefit from this rich and well-established theoretical literature. This article draws from that tradition to strengthen the mechanistic foundation of dark patterns research. The selection of theories reviewed in this article did not follow a systematic approach. The aim is not exhaustive coverage, but to foreground influential and broadly applicable consumer and decision-making theories that enable psychologically mechanistic explanation and testable predictions in persuasion-oriented dark patterns research. Therefore, I focus on a set of general, widely applicable consumer behavior theories that have strong explanatory and predictive value. I do not imply universal applicability of consumer theories to all domains where dark patterns are used, but rather, discuss general consumer decision-making theories that are applicable to a variety of contexts in which consumers' decisions are the focal analysis, many of which have a focus in the commercial domain, and I discuss more focused theories applicable specifically to persuasive technologies and deceptive practices. A comprehensive review of all potentially relevant consumer theories is beyond the scope of this paper, but the theories included here are representative starting points. Readers interested in more domain-specific or granular theories are encouraged to consult the sources that cited the reviewed literature. Overall, I argue that dark patterns research can draw from a wide and diverse range of predictive theories (which introduce specific measurable psychological constructs) that support rigorous hypothesis testing and mechanistic understanding of dark patterns, e.g., to identify the most and least harmful dark patterns.

The literature on dark patterns has developed several taxonomies and ontologies to define the different types of dark patterns and some of the psychological mechanisms through which they work (e.g., Gray et al., 2024; Jarovsky, 2022; Mathur et al., 2019). The ultimate aim of this article is to equip dark patterns scholars with a broader theoretical toolkit for understanding how and why digital designs influence consumer behavior—and how those influences may vary across people, contexts, and outcomes. The theoretical frameworks described here should help with a refined development of dark patterns taxonomies and ontologies in the future, and be an impetus for further, more specific, theory-development with a focus on dark patterns. It is not the aim of this article to refine such taxonomies or ontologies or existing theories, but to provide a general theoretical basis for such a refinement. Because I assume that readers are familiar with main dark pattern classifications (Gray et al., 2018, 2024; Mathur et al., 2019), I do not revisit those definitions in detail.

To identify the relevant theories and frameworks for this paper, I employed a search strategy combining various approaches. I used search terms such as “consumer psychology theory,” “consumer psychology framework,” “consumer theory,” “consumer framework,” “behavioral model,” “decision model,” and “decision theory” (and any combination thereof and related terms) to query Google Scholar. These terms guided my exploration across academic databases, ensuring a comprehensive review of literature pertinent

to consumer behavior and decision-making processes. Additionally, I examined theories referenced in Chang et al. (2024), which was the first paper to highlight theoretical gaps in dark patterns research. Beyond these searches, I integrated theories and models I was familiar with through my prior literature reviews and research. This multifaceted approach enabled me to select theories I found most relevant and applicable to dark patterns scholarship, focusing on choice architecture, decision-making in general (applicable to digital environments), heuristics, biases, and consumer vulnerabilities.

2. Heuristics, biases, and the dual-process model are not enough

Dark patterns research has often relied on psychological concepts related to rationality and decision quality, particularly the idea that dark patterns exploit heuristics—cognitive shortcuts that enable fast, efficient decision-making but can lead to systematic errors or biases (Dowling et al., 2020; Mildner et al., 2024; Oaksford & Hall, 2016; Todd & Gigerenzer, 2012). The heuristics-and-biases tradition, rooted in social and cognitive psychology, has documented how such shortcuts *may* produce judgments that deviate from normative—often deemed “rational”—decisions (Cialdini, 2021; Gigerenzer, 2002, 2004; Mildner et al., 2024; Stanovich et al., 2008; Todd & Gigerenzer, 2012)—and this is why they are often perceived or referred to as irrational. Researchers explain the occurrence of biases with the help of the dual-process model (Stanovich et al., 2008). The model suggests that humans have two systems of thinking. System 1 is fast, automatic, and intuitive, and does not require great cognitive capacities. This system is also described as experiential and affective. However, it is said to be prone to cognitive biases—i.e., faulty judgments, perceptions, and conclusions. System 2, on the other hand, is slow and effortful, and requires deliberation and attention, and it therefore is presumably much less prone to biased decision-making. Some researchers suggested that biases arise when System 2 decision-making is either not successfully executed or does not override the initial—potentially—biased response of System 1 decision-making. However, System 2 may also lead to biased, or erroneous decisions, especially when it is relying on statistical artifacts or false memory. Importantly, the dual-process model provides no all-encompassing theory that would allow mechanistic predictions of the multitude of dark patterns on consumer choices.

Despite their appeal, these models provide limited theoretical traction for predicting when and how specific dark patterns will influence behavior. The dual-process model lacks a detailed account of when System 1 vs. System 2 is activated, under what conditions one overrides the other, and how specific contextual cues interact with individual traits to drive decision processes. It does not allow us to make empirically precise predictions about which users are more vulnerable to specific dark patterns, why, or in which digital environments. It also offers little guidance on whether particular interface features (e.g., button prominence or message framing) are more likely to elicit intuitive vs. deliberative responses—and what consequences this might have for decision quality. While cognitive biases (e.g., scarcity, anchoring, default effects) describe behavioral tendencies, they do not constitute theories in the formal sense. They do not specify mechanisms that link design features to outcomes, nor do they offer generalizable predictions across contexts. As a result, dark patterns research that relies only on heuristics and biases often remains descriptive rather than predictive or explanatory.

3. Dark patterns may harm (vulnerable) consumers

Consumer vulnerability is a central concern in dark patterns research (Abbott et al., 2023; Luguri & Strahilevitz, 2021; Zac et al., 2024). While existing work has rightly emphasized cognitive mechanisms—such as heuristics and biases—this focus has often overlooked the broader range of factors that influence consumer behavior. Established theories in consumer psychology highlight the importance of affect, motivation, goals, personality traits, and contextual constraints. These and other dimensions can help explain why different dark patterns affect different consumers in different ways.

To move toward testable, mechanistic explanations of dark pattern effects, researchers should draw from comprehensive theories that go beyond information processing. Consumer behavior theories allow for predictions based on the interaction between individual characteristics and features of the choice environment (Burton, 2005; Sutton & Staw, 1995). It is plausible that different types of dark patterns exploit different mechanisms—and thus require different theoretical bases. These theories may also help

dark patterns researchers to form more specific theories that explain why different dark patterns have a differential influence on consumer choices. And they may help some scholars in refining the ontology of dark patterns (Gray et al., 2024). In the sections that follow, I briefly introduce a selection of such theories in order of their publication year,⁴ including the Stimulus-Organism-Response (S-O-R) theory (Jacoby, 2002), the capabilities-opportunities-motivation-behavior (COM-B) model (Michie et al., 2011; West & Michie, 2020), and the model of goal-directed behavior (Bagozzi, 2024), to name a few. I also highlight frameworks specific to deception in digital environments (Xiao & Benbasat, 2011) and persuasive design (Fogg, 2009), as well as theories relevant specifically to consumer vulnerabilities (Jacoby, 2002; Mowen, 2000b; Oreg & Bayazit, 2009). I relate each theory to dark patterns research and synthesize the theories, in hope of providing an overview of relevant theoretical foundations for dark patterns researchers to form empirically testable hypotheses that allow to make mechanistic predictions; to develop more nuanced theories for dark patterns; and to eventually extend the ontology of dark patterns with an understanding of the mechanisms that play a role in each type of dark pattern. Following the introduction of each theory, I provided a synthesis of these theories with a table overview of each theory's strengths, limitations, key concepts, and how it might be applicable to dark patterns research.

4. Ajzen's (1991) Theory of Planned Behavior

The Theory of Planned Behavior (TPB; Ajzen, 1985, 1991) explains human behavior within specific situations as driven by behavioral intentions, which are shaped by three factors: attitudes toward the behavior, subjective norms, and perceived behavioral control. These constructs can be meaningfully linked to how dark patterns influence consumer actions.

Intentions indicate how hard people are willing to try and how much effort they plan to exert to perform a behavior. The stronger the intention, the more likely the behavior will be performed. Ajzen emphasized that intentions will only predict a behavior if a person can perform this behavior—and this depends on the availability of opportunities and resources (e.g., time, money, skills, cooperation of others) and individuals' attitudes and norms that guide their behavior. The availability of opportunities and resources is defined as people's "actual control over the behavior" (Ajzen, 1991, p. 182). In this regard, Ajzen introduced the concept of *perceived* behavioral control—representing an individual's perception of the ease or difficulty of performing a target behavior, which is strongly related to the concept of self-efficacy, i.e., the belief in oneself that one *can* execute an action successfully (Bandura, 1982, 1991). Perceived behavioral control also reflects past own and peers' experiences and perceived obstacles to execute a target behavior. This perceived behavioral control (or self-efficacy) is thus influenced by the perceived chance of succeeding in the target behavior, which could be directly influenced by dark patterns, such as when consumers have a hard time understanding the platform due to the symbols or wording used (e.g., as in *Alphabet Soup* and *Untranslation* dark patterns). For example, Interface Interference or Language Inaccessibility patterns reduce users' sense of control by hiding options, using ambiguous language, or overloading the interface, even when the technical option remains available. This may discourage intended behaviors like unsubscribing or rejecting cookies.

The TPB posits that perceived behavioral control and intentions directly and independently predict target behaviors. The TPB is applicable to dark patterns research as they define the situational context within which a behavior should occur. According to the TPB, then, specific dark patterns might influence behavioral intentions and/or the perceived behavioral control to perform an action. Some dark patterns intentionally create obstacles to performing an action, such as is the case with the Interface Interference or Language Inaccessibility dark patterns. When consumers encounter unclear wording, hidden buttons, or overloaded settings, they may perceive less control over their actions—even when those actions remain technically available. This is a theoretical mechanism by which some dark patterns might influence behavior that has not been explored thoroughly yet. Intentions and perceived behavioral control are directly influenced by 1) attitudes toward a behavior and 2) subjective norms. 1)

Attitudes reflect the degree to which a person has a favorable or unfavorable emotional (affective) or cognitive evaluation of the behavior. Attitudes are formed based on beliefs about the outcomes of the behavior and the value placed on these outcomes. Attitudes can be very specific to an individual and are highly influenced by the individuals background and experiences. Here, especially experiences of

similar situations are relevant, which have resulted in desirable or undesirable consequences. It has been shown that dark patterns might influence the emotional state of consumers, thereby influencing their choices. For example, Luguri and Strahilevitz (2021) demonstrate that dark patterns such as confirmshaming, obstruction, and forced continuity elicit feelings of guilt, pressure, and anxiety, which increase acquiescence to unwanted choices. Similarly, Löschner and Pannasch (2024) show that deceptive interface designs can induce negative affect and emotional discomfort—such as frustration or stress—which impair reflective processing and bias subsequent decisions. More research on the effects of dark patterns on attitudes would thus be highly welcome to better understand the mechanisms that drive behavior via dark patterns.

Subjective norms refer to the perceived social pressure to perform or not perform a behavior and are influenced by normative beliefs regarding what others think about the behavior and the motivation to comply with these perceived expectations. Some dark patterns rely on the concept of *Social Proof* (Gray et al., 2023; Mathur et al., 2019), that is, the fact that people are heavily influenced by other people's opinions and actions within a given situation (Cialdini, 2021; Melnyk et al., 2022). Therefore, according to the TPB, dark patterns that leverage the concept of *Social Proof* directly influence perceived norms and thereby may indirectly affect intentions and perceived behavioral control, and through this mechanism affect consumer behaviors. Whether this mechanism is operating in dark patterns—and in which dark patterns specifically—as posited here, is partially an open question (much consumer research has examined the influence of social proof/norms, however, so dark patterns researchers might draw from this body of knowledge; see Albarracín et al., 2024; MacCoun, 2012; Melnyk et al., 2022).

Ajzen emphasized that the relative influence of attitudes, norms, and control varies by context (Ajzen, 1991, p. 188)—a useful feature for dark pattern research, which deals with highly situational digital environments. TPB thus allows us to form testable hypotheses about how specific design features influence behavior via these psychological pathways. For example, TPB would predict that interface interference reduces opt-out behavior by lowering perceived behavioral control, whereas social-proof-based dark patterns increase compliance by strengthening perceived subjective norms. Therefore, some dark patterns might be more or less influential, which likely depends on the target behavior that is affected by the employed dark pattern(s). For example, if a website is designed in such a way that it is difficult to find specific buttons that lead to a consumer-intended behavior, as is the case with dark patterns relying on *Interface Interference*, the consumer-intended behavior will occur less likely, and, instead, the website-intended behavior will occur more likely.

Digital platforms may use the constructs of TPB to gauge users' intentions toward behaviors such as signing up for a newsletter, making a purchase, or sharing content. By measuring attitudes, subjective norms, and perceived control, platform designers can better understand what drives users' choices. For example, by emphasizing positive outcomes of desired behaviors (e.g., benefits of signing up), websites can influence users' attitudes. This could involve testimonials or highlighting *Social Proof* to enhance subjective norms. Perceived behavioral control could involve making processes more cumbersome to make it harder for users to complete vendor-undesired actions, such as when consumers wish to unselect cookies, by employing *Obstruction* or *Interface Interference* types of dark patterns. Furthermore, providing consumers with feedback on their actions or support to overcome obstacles (e.g., FAQs or customer service) can improve their perceived control and thus increase the likelihood of desired behaviors being performed—the opposite is true when feedback is lacking or made very difficult to understand as with *Language Inaccessibility* dark patterns. By linking the psychological drivers of intention and control to specific design features, TPB provides a mechanistic, empirically grounded lens for understanding the impact of dark patterns across contexts. It can thus support both hypothesis-driven research and ontology refinement by enabling dark patterns to be classified according to the psychological pathways they activate (e.g., patterns targeting perceived control, subjective norms, or attitudes), rather than only by surface-level interface tactics.

5. Mowen's (2000) Meta-Theoretic Model of Motivation and Personality

The Meta-Theoretic Model of Motivation and Personality, referred to as the 3M model (Mowen, 2000a, 2000b), aims to provide an integrated framework that connects various psychological theories and

constructs related to motivation and personality, with the aim of providing one parsimonious framework that can explain consumer behaviors in different contexts. Mowen identified that many traditional theories of motivation and personality (e.g., those by Maslow, McClelland, and Freud) have been criticized for their limited empirical support and their inability to account for significant variance in behavior. The 3M model integrates elements from control theory that provides a framework how personality traits function to guide behavior (Carver & Scheier, 1990; Hyland, 1988), evolutionary psychology that examines the origins and the adaptive nature of personality traits (Buss, 1991); and the hierarchical structure of personality traits (Ashton et al., 2004; Markon et al., 2005; Paunonen, 1998) to provide a unified, mechanistic view of behavior. This makes it especially relevant for dark patterns research concerned with individual differences in vulnerability and context-specific behavioral tendencies. To account for the hierarchical structure of personality and individual differences, the 3M model is structured with four levels of traits:

- **Elemental Traits:** These are the most basic traits resulting from genetics and early learning. Mowen identified eight elemental traits, including Big Five personality traits (which were hence extended to the HEXACO personality traits; see Ashton & Lee, 2007; John, 2021; Lee & Ashton, 2004), material needs, physical/body needs, and the need for arousal. These traits predict behaviors across various domains (Beck & Jackson, 2022) and may be useful in predicting consumer choices (Matz & Hirsh, 2020).
- **Compound Traits:** Formed by combining elemental traits with cultural influences and individual learning experiences, these provide a focused view of personality that is still broad enough to encompass various behaviors. For example, compound traits reflect an individual's tendency to set high expectations for how a task is to be performed and to engage in the process of cognitive appraisal, e.g., influencing consumers' approach to shopping and product selection.
- **Situational Traits:** These are traits expressed in specific contexts, influenced by the situation (e.g., health, sports interest). In personality psychology, these are related to personality states, i.e., the manner in which a person expresses a specific trait within a given situation (Fleeson & Jayawickreme, 2015; Steyer et al., 2015) such as when shopping or choosing privacy settings online.
- **Surface Traits:** These represent enduring domain-specific behavioral tendencies (e.g., healthy eating, compulsive buying).

The model employs a control theory framework, where a “comparator” (i.e., the consumer) assesses outcomes against the reference values established by these traits, leading to cognitive appraisals that guide behavior. The integration of these components allows the 3M model to explain how underlying personality traits influence motivation and behavior in various consumer contexts. As such, this model may be useful in dark patterns literature that examines consumer vulnerabilities from a personality psychology perspective. Several scholars have pointed out that personality traits may be exploitable factors on digital platforms. For example, messages tailored to psychological profiles can affect participants' attitudes toward and purchase intentions of specific products (Hirsh et al., 2012; Moon, 2002; Wheeler et al., 2005) and even lead to higher conversion rates (Matz et al., 2017). Research thus suggests that the vulnerable consumer could be potentially identified by personality traits (Abbott et al., 2023) and other individual characteristics (Rossi et al., 2024).

The theory suggests that personality traits may moderate dark patterns' effectiveness as personality traits influence how people process information differently. This trait-based structure enables the formulation of testable hypotheses, such as “*Consumers high in emotionality will show stronger purchase intent in response to limited-time offers, compared to those low in emotionality*” or “*Those high in materialism or need for arousal may be more reactive to Scarcity or Social Proof tactics.*” Moreover, 3M highlights the importance of surface and situational traits in specific digital environments (e.g., online shopping, app subscriptions, privacy settings), enabling refined predictions for dark pattern effects in domain-specific contexts. It also supports ontology development by identifying trait-specific vulnerabilities to particular pattern types, helping explain heterogeneous behavioral effects across users. Overall, Mowen's 3M model offers a mechanistically grounded, trait-sensitive framework for studying how personality shapes susceptibility to dark patterns.

6. Jacoby's (2002) Stimulus-Organism-Response Theory

Jacoby's (2002) Stimulus-Organism-Response (SOR) Theory models consumer behavior as a dynamic interaction between external stimuli (S), internal mental states (O), and observable responses (R). It was proposed to overcome the limitations of overly rational or linear consumer models by acknowledging that behavior is mediated by complex internal processes. The rationale behind this theory is the recognition that consumer behavior is influenced by a multitude of dynamic, interactive factors that cannot be adequately represented by rigid, static models. In particular, the author argued that consumer rationality is an outdated assumption that minimizes individual differences “to the point of irrelevancy” (Jacoby, 2002, p. 51), which suggests that “irrationality” is the main component to be accounted for in complex consumer behavior models. The model thus is relevant to dark patterns scholarship that often argued that dark patterns exploit consumer irrationalities that stem from heuristic decision-making and cognitive biases (Mildner et al., 2024; Waldman, 2020). However, going a step further, the SOR framework emphasizes the importance of the internal mental processes of consumers (the “Organism”) that mediate or moderate responses to external stimuli. SOR theory introduced the following three main components:

- **Stimulus (S)** represents the external factors that impact consumer behavior, such as product information, interface elements, or dark pattern features. It encompasses all that the consumer encounters in their environment at a particular moment in time. S may thus represent the various dark patterns employed on digital platforms.
- **Organism (O)** refers to the individual differences in the consumers' internal processes, including cognitive and emotional responses, beliefs, attitudes, memories, and personality traits. The organism is responsible for processing the stimuli and determining the responses based on its internal state. The organism hence is the lens through which the external stimuli are interpreted; the factors representing the organism thus may act as moderators of the stimulus (see Abbott et al., 2023 for a similar argumentation, applying the SOR theory to impulsive online shopping in a dark patterns study). For instance, users with high impulsivity may respond more strongly to limited-time offers, while those with negative digital experiences may interpret unclear buttons as deceptive.
- **Response (R)** encompasses the observable behaviors or reactions of the consumer following the internal processing of stimuli. Responses can be both internal (like changes in attitudes or satisfaction) and external (like webpage/app clicks, purchases, abandoning a process, or engaging in communication).

Instead of displaying S, O, and R sequentially, Jacoby suggested representing them as overlapping circles in a Venn diagram, which indicates the dynamic interactions among these components and highlights the permeable boundaries between them. Jacoby thus explicitly added a temporal dimension, emphasizing that stimuli and organismic states interact differently across time (e.g., pre- vs. post-purchase), which could help explain delayed regret or post-transaction complaints—a largely unexplored area in dark patterns research. This illustration also accounts for the argument laid out by Jacoby that it often is unclear whether one of the factors is an input or output factor due to feedback loops. Displaying SOR theory in a Venn diagram led Jacoby to identify seven “sectors” that categorize different types of stimulus factors, organismic factors, and response factors (Figure 1 in Jacoby, 2002, p. 54), which aim to provide a comprehensive understanding of consumer behavior. This categorization assists in identifying and visualizing the complex interrelations among various factors influencing consumer behavior. These include

1. The **Encountered Environment (Sector 1)** includes all external stimuli encountered by the individual at a particular moment, such as the displayed products, brands, ads, and dark patterns.
2. **Automatic Processing (Sector 2)** involves subconscious processing of stimuli, where certain stimuli may be processed without conscious awareness such as is the case with implicit learning and automatic interpretations of stimuli (light, smells, temperature). This sector is probably most related to heuristic decision-making and cognitive biases and hence may play a role in some dark patterns that specifically exploit heuristics and biases (e.g., *Scarcity*, *Social Proof*).

3. The **Experiential Storehouse (Sector 3)** is the long-term memory storehouse, containing prior experiences, knowledge, beliefs, attitudes, predispositions, intentions, values, cognitive networks, schema, scripts, motives, the individual's personality, feelings, impressions, images, expectations, etc. It serves as the repository for all past interactions and learnings. This sector includes all individual differences that may be exploited to identify and target vulnerable consumers.
4. **Consciousness (Sector 4)** represents the individual's awareness of the moment, including active motives, moods, perceptions, and cognitive processes. This sector thus encompasses all factors that help consumers to evaluate a digital platform and may provide a theoretical basis for dark patterns effects based on faulty reasoning or information processing, both of which may lead to cognitive biases.
5. **Nontrace Stimulus-Response Events (Sector 5)**: This consists of automatic responses that do not register psychologically, such as reflexive behaviors or changes in behavior due to unnoticed stimuli. It seems appropriate to relate Sector 5 to cognitive biases, which result from unconscious processes based on often unnoticed stimuli (e.g., the scarcity bias is triggered by stimuli such as a limited availability of a product offer and consumers rarely notice that this stimulus has activated a process that led them to evaluate the product differently merely because a product was advertised as scarce).
6. **Internal Responses (Sector 6)**: This includes internal changes resulting from conscious processing, such as shifts in beliefs, attitudes, and intentions that are not directly observable. This sector is related to System 2 thinking—effortful, deliberate cognitive processes—which may still lead to wrong conclusions if the reasoning or “calculation” was faulty. If dark patterns influence beliefs, attitudes, and intentions, this sector allows for a more mechanistic understanding of which of these are influenced and why.
7. **External Responses (Sector 7)**: This encompasses all observable behaviors, including physical and physiological responses (blushing, sweating, pupil dilations, smiles, eye fixations), forms of communication (groans, sighs, writing, speaking), as well as actions like purchasing, using, or abandoning products/services. Past behaviors are generally a good predictor of future behaviors such that this sector provides a theoretical understanding for the factors that could be measured for future predictions of consumer decisions.

Together, these seven sectors aim to provide a comprehensive framework for understanding the dynamics of consumer behavior by acknowledging the interactions between external stimuli, internal cognitive processes, and observable responses. The SOR theory thus is related to Mowen's 3 M model, but SOR theory puts less emphasis on the personality traits that lead to the organism's different processing.

Jacoby explicitly introduced a temporal dimension to analyze consumer behavior, which refers to the understanding that consumers' motives, moods, attitudes, and responses can change between different points in time (e.g., from pre-purchase to post-purchase stages). This dynamic view helps account for how repeated exposures to stimuli and evolving internal states influence consumer decisions. While dark patterns research has traditionally focused on static interface elements, the importance of temporality has recently gained attention. For instance, Gray et al. (2025) proposed a methodology for Temporal Analysis of Dark Patterns, which maps dark patterns across intra-page, inter-page, and system-level interactions to capture their cumulative effects over a user's journey. Although their framework does not draw from consumer theories, it operationalizes similar insights to SOR theory—namely, that user behavior cannot be fully understood without considering how stimuli unfold and interact over time. Integrating Jacoby's temporal lens into dark patterns research may thus enhance both theoretical depth and empirical rigor.

7. B.J. Fogg's (2009) Behavior Model for Persuasive Technologies

B.J. Fogg's (2009) Behavior Model for Persuasive Technologies (FBM)⁵ provides a framework based on insights from psychology to understand consumer behavior in the context of persuasive design. This model, therefore, maps closely to how some dark patterns manipulate user action—as these are persuasive (deceptive) designs that leverage humans' cognitive biases (psychology) on digital platforms.

According to FBM, a target behavior occurs when the following “three principal factors” converge; otherwise, the behavior will not happen:

- **Motivation** refers to the consumers’ desire to perform the target behavior. Motivation can vary in intensity and is influenced by three core motivators: immediate reactions to situations based on pleasure/pain; the anticipation of outcomes resulting in hope/fear, and the human desire for social acceptance and fear of social rejection. High motivation is essential for a target behavior to occur. The three core motivators are related to heuristics and biases (e.g., *Loss Aversion* distinguishes between different responses in pleasurable/rewarding and fearful/disappointing situations) and to dark pattern types (e.g., *Social Proof* is partially based on the desire to be socially accepted, and the *Scarcity Bias* is partially driven by fear of missing out on an offer).
- **Ability:** This denotes the individual’s capacity to perform the target behavior (e.g., the purchase of a product), which can include skills, resources, and ease of performing the task. Simplicity is key: the easier a behavior is to perform, the more likely it is to happen. Some dark patterns (e.g., Obstruction, Hard to Cancel, and Interface Interference) manipulate the ease of an action by making undesired behaviors (e.g., unsubscribing, rejecting cookies) harder, thus reducing perceived ability. As an example, Fogg argued that features like 1-click purchasing demonstrates how simplifying the process can lead to increased target behaviors (purchases). Fogg identified six elements (resources, skills) that contribute to a person’s ability to perform a behavior: (1) the time required to perform the behavior; (2) the financial cost associated with the behavior; (3) physical effort required to perform the behavior; (4) the cognitive effort required to perform the behavior (System 1 versus System 2 thinking); (5) whether the behavior conforms with or goes against established social norms (*Social Proof*); and (6) whether the behavior is within or outside of an individual’s routine as non-routine behaviors are perceived as less simple. Ability is also affected by contextual factors—the circumstances and context within which a behavior is supposed to occur—and is limited by the scarcest resource available—e.g., if the consumer does not have the time, financial means, physical or cognitive capacity to perform a behavior, it will not occur. The ability component in the FBM emphasizes the importance of reducing barriers to behavior performance through simplicity. All of the factors mentioned here are measurable and could be used as variables in dark patterns research to statistically test FBM’s predictions.
- **Triggers** (a.k.a cues or prompts) are necessary signals that prompt the individual to engage in the behavior at the right moment. Without an appropriate trigger, even high motivation and ability may not lead to the desired behavior. On the other hand, a trigger will only lead to the desired behavior if the consumer has sufficient motivation and/or ability to perform the behavior. Triggers have three characteristics to be successful: 1) they must be easily noticeable, 2) they must be associated with a target behavior, and 3) they must be active at a time the consumer is both motivated and able to perform the target behavior. Fogg identified three main types of triggers: (1) Sparks are designed to motivate behavior, such as with emotionally charged texts (like in the *Toying With Emotions* dark pattern) or messages that highlight fears (like in *Scarcity* messages). (2) Facilitators help make a behavior easier to perform, such as with a 1-click option to complete a process quickly. Once facilitators were identified, these could, however, also be used against the consumer as in the *Hard to Cancel* dark pattern. (3) Signals serve as reminders for behaviors when individuals already possess both the motivation and ability to act. Signals indicate when a behavior is appropriate. Such signals can, however, be used against the consumer. For example, a green color typically elicits a “go for it” behavior, whereas a red color indicates a “stop, be careful” behavior. Interface designers could exploit these tendencies by making the “Opt out” button red and an “Opt in” button green when triggering a consumer to respond to a subscription.

The FBM emphasizes that for a behavior to occur, the three components—motivation, ability, and an effective trigger—must be present simultaneously. Fogg notes that the FBM can be used to design systems that prevent target behaviors to happen—such as with *Obstruction*-types dark patterns. If any one of these factors is missing, the behavior will not happen. This prediction is empirically testable when all three components are measured within the same study design. This model thus provides a

systematic approach for designers and researchers to analyze and design persuasive technologies by addressing these three critical elements, helping them identify factors within those three components that could affect consumer choices. Though FBM is not designed to explain individual vulnerabilities, it helps specify the immediate conditions under which a dark pattern succeeds or fails. This makes it a strong candidate for refining ontologies by categorizing patterns according to whether they manipulate motivation, suppress ability, or inject deceptive triggers. The model does not provide a thorough theoretical basis to understand consumer vulnerabilities; but researchers need not apply only one theory and the following theory could be one such theory appropriate to hypothesize consumer vulnerabilities in dark patterns research.

8. Oreg's and Bayazit's (2009) Bias Taxonomy from an Individual Differences Perspective

The Bias Taxonomy from an Individual Differences Perspective (Bias Taxonomy) seeks to explain how various cognitive biases arise from the interplay between individual differences and motivational goals (Oreg & Bayazit, 2009). Unlike models that treat biases as universal errors, this taxonomy emphasizes that people differ systematically in their susceptibility to specific biases—making it highly relevant for dark patterns research, which increasingly examine consumer vulnerabilities (Abbott et al., 2023; Rossi et al., 2024; Zac et al., 2024). The Bias Taxonomy provides a framework for understanding such consumer vulnerabilities and allows to derive empirically testable hypotheses. The Bias Taxonomy also allows to better understand dark patterns that rely on cognitive biases, such as *Scarcity*. The framework categorizes biases into three functional types, each rooted in distinct motivational and personality mechanisms:

Verification biases relate to individuals' perceptions of themselves and their environments, often leading to self-enhancement or self-depreciation. They arise from the motivation to achieve consistency and coherence in self-concept and may lead individuals to distort information about themselves or their environment. These biases may explain responses to Confirmshaming, where self-image is manipulated to steer compliance.

Simplification biases arise from the need to reduce cognitive effort when processing complex stimuli. These underpin heuristics and shortcuts and are likely activated by dark patterns like Nagging, Forced Continuity, or Preselection—where repeated exposure or interface friction leads to compliance without deliberation.

Regulation biases pertain to decision-making processes in contexts involving risk and uncertainty. They stem from the basic human motivations to approach pleasure and avoid pain. These biases can explain behavior when consumers are exposed to dark patterns that exploit scarcity, urgency, or social proof cues, where fear of missing out or the desire for acceptance overrides deliberation.

The taxonomy links these bias types to individual differences in cognitive style, core self-evaluations, temperament, and motivation. For instance, consumers with lower cognitive capacity or individuals with distorted perceptions as measured by the disintegration/psychoticism personality trait (Ashton & Lee, 2020; Knežević et al., 2017) may be more prone to simplification biases; those high in neuroticism or low self-esteem may be more susceptible to verification or regulation biases. These individual differences interact with motivational goals—such as the need for autonomy, control, or social inclusion—to shape how information is processed and which biases are most likely to manifest.

Crucially, the theory posits that biases are consequences of individuals' attempts to satisfy basic motivational goals—so-called motivational end states that reflect deeper psychological needs. This makes the model particularly valuable for dark patterns research: dark patterns can both trigger motivational goals (e.g., urgency exploiting fear of loss) and leverage individual traits that modulate bias strength. This interaction allows researchers to develop mechanistic hypotheses about which users are more likely to comply with which patterns, under what conditions, and test these empirically. The insights from the Bias Taxonomy could help refine the ontology of dark patterns by providing a basis of the consumers' psychological make-up that helps to understand differential effects of these patterns. This framework also supports ontology refinement by offering a psychological logic for classifying dark patterns—not just by their surface features or targeted outcome, but by the bias mechanisms they exploit. Incorporating verification, simplification, and regulation biases into ontological structures could

improve the granularity and diagnostic value of dark pattern taxonomies, helping scholars distinguish between manipulative or deceptive strategies based on their underlying psychological triggers.

9. Xiao's and Benbasat's (2011) Theory of Product-Related Deceptive Information Practices in E-Commerce

Xiao's and Benbasat's (2011) Theory of Product-Related Deceptive Information Practices in E-Commerce (DIP-Theory) provides a framework for understanding how online merchants can deceive consumers through various manipulations of product-related information. The authors defined "product-related e-commerce deceptive information practices as the deliberate manipulation of product-related information perpetrated by online merchants to mislead consumers in order to induce desired attitudinal and behavioral changes in consumers—changes that are detrimental to consumers and beneficial to the merchants" (Xiao & Benbasat, 2011, p. 172) and thereby argues along the lines of dark patterns scholars (Bösch et al., 2016; Mathur et al., 2019; Narayanan et al., 2020; Voigt et al., 2021). By understanding how and why consumers are deceived, the theory aims to enhance consumer awareness, aid in the detection of deceptive practices, and inform the development of better regulatory and business practices—and thereby, this theory may be useful to dark pattern scholars, since dark patterns are deceptive and partially regulated.

The DIP -Theory introduces a 3×3 typology that categorizes deceptive practices into three main types: 1) *Concealment*: Withholding or omitting relevant information (e.g., negative product features) which is related to *Hidden Information* dark patterns; 2) *Equivocation*: Providing vague or ambiguous information (e.g., unclear return policies) which is related to *Obstruction* and *Interface Interference* dark patterns; 3) *Falsification*: Presenting false or exaggerated information (e.g., selling products as if they were scarce) which is related to *Social Engineering* dark patterns. These three types are further expanded upon in a second dimension of three specific techniques by which these types of deception can be implemented in e-commerce: First, by manipulation of information content—a direct alteration of the content of product information, e.g., omitting negative reviews, thereby aiming to avoid the negative influence of *Social Proof*. Thereby, digital platforms can deceive consumers into unrealistic beliefs and expectations. Second, by manipulation of information presentation, e.g., changing how product information is presented. This may involve using misleading visuals like in the *False Hierarchy* dark pattern or structuring a process such that it makes it difficult for the consumer to find a way to opt out of a service, which is a technique that is related to *Interface Interference*, *Misdirection*, and *Obstruction* dark patterns. With such techniques, digital platforms can direct consumers' choices toward the products/services/options that are most useful or profitable to the platform provider. Third, by manipulation of information generation, e.g., dynamically producing product information based on consumer preferences, often in a biased manner, perhaps by using skewed recommendations from algorithms, digital platforms can target specific consumers, which essentially makes consumers more vulnerable on digital platforms. This vulnerability has been highlighted repeatedly in the dark patterns literature (Schilling & Sergeeva, 2025; Schilling & Turrey, 2025) and the persuasive technology literature (Kaptein et al., 2015).

The theory also considers how these deceptive practices affect consumers' cognitive and affective (emotional) responses, and how individual and situational characteristics influence consumers' judgments about product value and ultimately their purchasing decisions. To this end, the theory employs the previously summarized Stimulus-Organism-Response Theory (Jacoby, 2002). DIP -Theory furthermore outlines the processes consumers may go through in detecting deception, such as 1) noticing anomalies, i.e., recognizing inconsistencies in product information and 2) attribution of anomalies, i.e., determining whether the observed inconsistencies are due to deceptive practices. According to DIP-theory, individual, product, and situational characteristics moderate how consumers perceive and respond to deceptive practices, thus affecting their vulnerability to deception. For instance, a consumer's prior trust in a website, product expertise, and the complexity of the shopping task can all influence their ability to detect deception. The authors point out that feelings of pleasure and arousal triggered by deceptive practices may lead to impulsive buying behavior, thereby tying neatly into dark patterns research, as one of the main goals of vendors employing dark patterns is to elicit impulsive buying behaviors (Abbott et al., 2023). DIP-Theory is furthermore related to dark patterns research through its

proposition that processing of (deceptive) information may undermine consumers' self-control and reduce cognitive attention—two concerns that have been raised in the dark patterns literature, too (Graßl et al., 2021; Mildner et al., 2024; Monge Roffarello et al., 2023).

Overall, the theory provides an integrative model that enhances the understanding of deception in e-commerce, offering a foundation for further research and practical applications in dark patterns research that is primarily concerned with consumer protection and business ethics. The theory identifies several moderators of susceptibility—including prior trust, product expertise, and task complexity (concepts of which some have not yet been mentioned in other theories or frameworks)—offering avenues for testable hypotheses about who is most vulnerable and under what conditions. The DIP-Theory is related to the other theories described within this article—Xiao and Benbasat specifically referred to SOR-Theory. DIP-Theory also builds on several other related theories that I have not summarized in this article, because they have either not been as influential (as determined by the citations) or I deemed them to be too specific so that they were less useful as a broad theoretical framework of dark patterns. One of these is the process-oriented model of deception detection (Johnson et al., 1993, 2001), which focuses on the cognitive processes that experts go through to detect deception after it has occurred—primarily in financial fraud and other low-frequency, high-stakes decision contexts—rather than on predicting how deceptive designs influence everyday consumer choices. For more details, I refer the interested reader to the article by Xiao and Benbasat (2011). DIP-Theory offers a behaviorally grounded, deception-specific framework that can guide hypothesis-driven research on dark patterns and help refine their classification. By mapping deception types and tactics to known dark pattern categories, researchers can move toward a functionally informed ontology that distinguishes dark patterns not only by design, but also by their mechanism of influence.

10. Xiao' and Nicholson's (2013) Multidisciplinary Cognitive Behavioural Framework of Impulse Buying

Xiao and Nicholson's (2013) Multidisciplinary Cognitive Behavioral (MCB) Framework of Impulse Buying presents impulse buying as a dynamic, multi-stage process shaped by psychological, emotional, and contextual factors. While the framework is not limited to digital environments, it is especially relevant for dark patterns research, as many deceptive design practices aim to elicit unplanned, emotionally driven purchases (Abbott et al., 2023; Luguri & Strahilevitz, 2021). The MCB Framework identifies four core components that together explain how, when, and why impulse buying occurs—each developed on the basis of its own theory:

1. Antecedents are the individual psychological characteristics and sociocultural factors that exist before the buying act, including personality traits (e.g., low conscientiousness, high neuroticism), beliefs (self-image, self-identity), values (materialistic values, social norms), and demographic factors (age, sex, income, educational background). These traits parallel vulnerability profiles already discussed in dark patterns literature and earlier models (e.g., 3M, Bias Taxonomy).
2. Triggers include the internal and external stimuli that activate impulse buying behavior. Triggers can be emotional states (activated by *Toying with Emotions*), situational cues (*Nagging*, *Obstruction*, *Interface Interference*), product presentations (again, *Interface Interference*), and promotional activities (*Social Engineering*) that provoke the impulse to buy. Triggers correspond to the “stimulus” in S-O-R theory and the “trigger” in FBM, and the framework emphasizes that affective and perceptual cues play a central role in prompting impulsive acts.
3. The act of buying refers to the actual purchasing behavior characterized by a sudden urge to buy, often with reduced cognitive evaluation due to emotional arousal and attentional focus on the product. Emotional arousal could be the result of *Toying with Emotions* dark patterns, and attentional focus could be the result of *Interface Interference* dark patterns. The act of buying is typically described in the literature by decision process theories, describing how inputs are processed into outputs. Xiao and Nicholson therefore highlighted emotional responses (e.g., System 1 responses), heuristic decision-making and information processing as important aspects during the (impulsive)

buying act. Hereby, they provided a link of their framework to dark patterns research that is based on heuristics and biases and the dual-process model.

4. Post-purchase outcomes encompass the emotional, cognitive, and behavioral reactions following the purchase. In the post-purchase phase, consumers evaluate their impulse purchase. The authors claim that the outcomes of the evaluation can influence future impulse buying, creating a cycle where prior purchases trigger new impulse buying behaviors or purchase aversion—and may eventually lead to compulsive buying behaviors. Furthermore, these outcomes may lead to regret and dissatisfaction, which ultimately may reflect negatively upon the vendor who lured the consumer into the impulsive purchase. From this perspective, dark patterns that lead to impulsive buying could also affect the vendor negatively, especially if the consumer complains about the vendor's techniques that made them buy impulsively. This has not been discussed clearly in any other theory and would provide a basis for exploring more the negative unintended consequences of dark patterns on the digital platform provider.

The MCB framework integrates perspectives from trait theory, regulatory focus theory, dual-process models, and S-O-R, making it both comprehensive and highly compatible with prior theories discussed in this article. It is particularly valuable for dark patterns research because it provides a temporal and emotional structure for understanding not only when dark patterns are effective, but also how they might backfire. As in other theories, the MCB framework again highlighted the interconnectedness of its components, and Xiao and Nicholson suggested that understanding impulsive buying requires considering how these elements interact to influence consumer behavior. The framework explains how (e.g., via antecedents) and when (e.g., via triggers) impulsive buying occurs, and what the emotional and behavioral consequences of impulsive buying are. Emotional responses can create a cycle where the post-purchase feelings either reinforce future impulse buying behavior or repel consumers from repeat purchases from the same vendor. These consequences highlight that impulse buying can be seen as both a process and an outcome.

What makes this article useful to dark patterns scholarship beyond other articles already referred to is that it introduces the post-purchase phase during which consumers evaluate their choices. This post-purchase phase can be critical to the consumer and the vendor so that those employing dark patterns better be aware of the potential negative consequences of dark patterns—irrespective of the fact that dark patterns have been started to be regulated and companies have been fined for their application of dark patterns (deceptive.design, 2024; Gray et al., 2023). The focus on emotions in this framework provides theoretical ground for future explorations of how dark patterns influence emotions (Löschner & Pannasch, 2024; Luguri & Strahilevitz, 2021; Tuncer et al., 2023) and the evaluation of online vendors.

Furthermore, this framework defined and described what impulsive buying is, based on a thorough literature review, so that this type of behavior is better understood and can be measured more precisely. The authors defined impulse buying as “an unplanned and sudden buying act, in response to subjective or external stimuli, accompanied by a powerful and persistent urge; after the purchase, the customer experiences emotional, cognitive and/or behavior reactions, which may become the new trigger of repeated [impulse buying]; a reflection of impulsivity traits, sociocultural values and buying beliefs; both as a process and an outcome” (p. 335). This definition can be refined to fit any impulsive consumer decision, such as newsletter/membership subscriptions or cookie consent acceptance.

In sum, the MCB Framework enables testable, mechanistic hypotheses about how dark patterns leverage emotional arousal and individual vulnerabilities to induce impulsive action. It also opens the door for ontology refinement, especially by clarifying the phases of impulsive response and the feedback loops that determine whether dark pattern tactics result in consumer retention or rejection.

11. Michie's and West's (2020) Behavior Change Wheel and COM-B Model

The Behavior Change Wheel (BCW) and the COM-B model (Michie et al., 2011; West & Michie, 2020) serve as frameworks for understanding and designing behavior change interventions—such as dark patterns that influence consumer choices. The BCW and COM-B models address the complexity of human behavior (B), which is influenced by a variety of factors, subsumed under capability (C), opportunity

(O), and motivation (M). Originally developed for public health and clinical practice, the model is also applicable to other decision contexts such as digital environments. The model aims to assist intervention developers in identifying appropriate targets for their behavior change strategies by linking theoretical constructs to practical applications. In this model,

- **Capability** refers to the psychological and physical capacities necessary for behavior, including knowledge, cognitive skills, and self-regulatory competence. This concept overlaps with user characteristics that dark patterns may exploit—such as limited attention, cognitive overload, or digital literacy gaps. Some users may lack the capability to resist manipulative prompts due to these constraints, linking directly to the concept of consumer vulnerability.
- **Opportunity** encompasses all external factors that facilitate or constrain behavior. Dark patterns often modify the digital environment to manipulate opportunity: e.g., Interface Interference, Obstruction, or Nagging may restrict visible alternatives or overload users with prompts, thereby reducing their ability to act in accordance with their preferences. This component aligns closely with digital “choice architecture” and explains how affordances are engineered to channel behavior.
- **Motivation** involves all the internal (cognitive) processes that energize and direct behavior, including reflective processes (calculations, evaluations, considerations) and automatic processes (habits, unquestioned values and beliefs, desires, and emotions). Some dark patterns hijack emotional responses, others exploit values and beliefs (e.g., social norms). The COM-B model’s inclusion of both System 1 (automatic) and System 2 (deliberative) processes offers a mechanistic link to dual-process theories of decision-making often invoked in dark patterns literature through the Motivation pathway.

The COM-B model posits that behavior occurs when an individual has the capability and opportunity to enact it and is sufficiently motivated to do so. In this regard, this model with its components is similar to Fogg’s FBM (2009). The BCW expands upon the COM-B model by presenting a visual framework that categorizes interventions aimed at addressing deficits in capability, opportunity, and motivation. The BCW addresses shortcomings of other behavioral models by incorporating the feedback loops that drive behavior over time, ideas similar to those presented in SOR-Theory (Jacoby, 2002) and the MCB framework (Xiao & Nicholson, 2013). At the center of the wheel is the COM-B system, which is encircled by nine intervention functions that can be utilized to address the identified factors. These functions include education, persuasion, incentivization, coercion, training, restriction, environmental restructuring, modeling, and enablement. Surrounding these intervention functions are seven policy categories that can support or enable the implementation of the interventions. These categories include communication/marketing, guidelines, fiscal, regulatory, legislative, environmental/social factors, and service provision. The framework, hence, attempts to be multidisciplinary and applicable to many decision-contexts, including consumer behaviors on digital environments.

Dark patterns researchers can use the COM-B model to categorize different dark patterns based on how they affect users’ capabilities, opportunities, and motivations. For example, a dark pattern that employs *Confirmshaming* may be analyzed for its manipulation of motivation and opportunity, limiting user choices while inducing guilt. By categorizing dark patterns according to their effects on user capability, opportunity, and motivation, researchers can create a more cohesive vocabulary that aligns with behavioral theories. This taxonomy can aid in distinguishing between different types of dark patterns and their intended manipulative strategies, thereby facilitating clearer communication among scholars, designers, and regulators. The behavior change wheel can also inform how to design interventions to mitigate the harmful effects of dark patterns. For example, the *Nagging* dark pattern involves persistent prompts that interrupt consumers’ actions, often leading them to comply with unwanted requests, like subscribing to a service. Applying the BCW allows designers to explore intervention functions such as restructuring the physical or social environment to minimize nagging behaviors. In sum, COM-B provides a mechanistic vocabulary to describe how dark patterns exert their influence and can aid scholars in categorizing, predicting, and potentially intervening in their effects. Integrating the COM-B model into dark patterns research helps bridge theoretical and applied perspectives and supports the refinement of dark pattern ontologies by identifying which behavioral mechanisms each pattern targets or exploits.

12. Bagozzi's (2024) Model of Goal-Directed Behavior

Similar to the argumentation on which the previous models and frameworks build (Jacoby, 2002; Mowen, 2000a; Xiao & Nicholson, 2013), Bagozzi (2024) argued that the literature is fragmented and needs to be consolidated in one comprehensive framework and that post-consumption behaviors and outcomes need to be accounted for, ideally in an ongoing outcome process. The MGDB addresses this gap by synthesizing psychological constructs (e.g., emotions, desires, attitudes), contextual factors (e.g., culture, norms, digital environment), and vendor characteristics (e.g., brand personality, community) into a unified model. Bagozzi's Model of Goal-Directed Behavior (MGDB) hence offers one of the most comprehensive and integrative frameworks for explaining consumer behavior, building on decades of empirical and theoretical work, including an earlier model extension of the Theory of Planned Behavior (Ajzen, 1991) introduced by Perugini and Bagozzi (2001). Bagozzi (2024) distinguishes between three analytical levels of theory—macro, meso, and micro—referring to different degrees of abstraction at which consumer behavior can be explained. While these levels serve a different purpose than Gray et al.'s (2024) multilevel ontology of dark patterns, both frameworks share a commitment to organizing complex phenomena across levels of abstraction. In particular, the meso level plays a central role in both approaches: Gray et al. use it to capture functional mechanisms underlying groups of dark patterns, whereas Bagozzi uses it to integrate mid-range psychological processes linking context, motivation, and behavior. This conceptual alignment at the level of abstraction—rather than a one-to-one correspondence of constructs—suggests that the MGDB can serve as a complementary explanatory framework for theorizing how different classes of dark patterns exert their influence. For example, a meso-level dark pattern category such as interface interference can be understood descriptively within Gray et al.'s ontology, while the MGDB specifies how such situational manipulations translate into desires, anticipated emotions, and intentions at the consumer level. This compatibility suggests that the MGDB is particularly well-suited to support theory development in dark patterns research.

The MGDB thus took the TPB as its core and extended it by consumer characteristics (e.g., past behavior, desires, emotions, attitudes, identity, personality, self-concept, brand reactions), situational characteristics (e.g., culture, norms, physical environment), and vendor characteristics (e.g., brand personality, brand community). Each of these characteristic categories can be further broken down as follows.

Consumer characteristics include past behavior (one of the best predictors of future behavior), individual differences (e.g., traits, values, identity, self-concept, attachment styles, heuristics), and desires. Desires are a central component of the MGDB and are differentiated into intrinsic, instrumental, goal, and action desires—each shaped by anticipated emotions, attitudes, values, and social norms. These desires provide the motivational impetus for intentions, unlike the TPB, which treats intentions as directly shaped by attitudes and norms without considering motivational depth.

Situational characteristics are most related to dark patterns, which are often argued to be methods of choice architecture (Chang et al., 2024), as these are external factors within which consumers make their choices. Situational characteristics include a) *culture* (Hofstede, 2001, 2011); b) *substantive manipulations* referred to as intentional changes in the variables or conditions of a study to observe their effects on consumer behavior; and the c) *physical (or digital)* environment, encompassing the surroundings and conditions in which consumer behavior occurs, influencing decision-making and emotional responses.

Vendor characteristics include factors such as a) *brand community* refers to the consumers targeted, who tend to share individual characteristics, an emotional connection and have a sense of loyalty toward the brand and b) *brand personality* is the set of human-like personality traits attributed to a brand, which influence consumer perceptions and emotional connections, making the brand relatable and distinctive.

Overall, the MGDB provides a highly integrated and dynamic understanding of consumer behavior, emphasizing the interplay between cognitive, emotional, and motivational processes in goal-directed actions (i.e., consumer behaviors), and the influence of individual differences and situations as moderators of these processes. Key differences in the MGDB compared to other theories and frameworks are that the MGDB introduced a nuanced definition of intentions: *I*-intentions such as distal, proximal, and motor intentions that address actions and behaviors of individuals versus *We*-intentions that allow to address collective actions and behaviors of groups of people (or couples). The MGDB also incorporates insights from neuroscience, linking neurochemical processes (molecules, hormones), emotional

responses, and cognitive control mechanisms that influence consumer decision-making. As some previous theories (e.g., Jacoby's SOR theory, Xiao' and Nicholson's MCB framework), the MGDB has a recursive structure and treats behavior as both input and output. Past behavior influences present desires and intentions, and post-behavior outcomes (e.g., regret, satisfaction) feed back into future decisions. This looped structure makes the MGDB particularly well-suited for theorizing about long-term effects of dark patterns, which operate through repeated exposure and subtle conditioning.

Dark patterns research can benefit from the MGDB in multiple ways. The model introduces a wide range of testable variables—both newly proposed and previously theorized—that can be empirically investigated in relation to dark patterns. For example, constructs like desires (intrinsic, instrumental, goal, and action), cultural orientations (individualism vs. collectivism), intention types (I- versus We-intentions), and neurobiological indicators (e.g., brain regions, hormones, neurotransmitters) offer new dimensions through which researchers can examine how dark patterns shape consumer decision-making. These variables are not only theoretically rich but also operationalizable in experimental and survey-based designs, making them suitable for hypothesis-driven empirical research. At the same time, the MGDB integrates variables found in prior models—such as individual differences in personality traits, values, identity, self-concept, and heuristics—but offers a more cohesive framework that situates them within a goal-oriented behavioral process. Notably, it is the first theory discussed here to explicitly include heuristics as a core basis for decision-making, bridging intuitive judgments with motivational and contextual moderators.

In sum, the MGDB provides a comprehensive, multi-layered framework for understanding the mechanisms through which dark patterns exert their influence. It allows scholars to formulate nuanced, mechanistically grounded hypotheses about which types of consumers are most susceptible, how manipulation operates across different contexts, and what longer-term outcomes may emerge. By integrating emotional, motivational, cognitive, social, and even biological variables, the MGDB offers dark patterns researchers a new and expansive lens to explore consumer vulnerability, manipulation, and resistance—thereby supporting both theoretical refinement and the expansion of dark pattern ontologies.

13. Synthesis of theories

The consumer behavior theories reviewed in this paper offer a diverse range of constructs that can guide theory-based dark patterns research that aims to empirically test hypotheses about these constructs. Collectively, the theories emphasize that behavior is shaped not only by the immediate digital environment (i.e., dark patterns themselves), but also by individual differences, motivational processes, emotional states, cognitive shortcuts, and broader social and cultural contexts. Only one of the theories discussed here—MGDB—incorporates heuristics and biases specifically as predictors of consumer behavior, suggesting that these have only a marginal influence on consumer choices. This starkly contrasts dark patterns research that prominently is based on heuristics and biases “theory.” Many summarized theories here overlap to some extent, yet each can inform the other as it adds distinctive variables to the explanatory landscape. These theories support mechanistic hypothesis development, allowing researchers to move beyond descriptive taxonomies of dark patterns toward predictive frameworks. For example, the TPB serves as a foundational framework that connects to other theories by emphasizing intentions as predictors of behaviors. The 3 M model integrates multiple theories related to motivation and personality to explain consumer behavior on a hierarchical level of individual differences (e.g., personality traits). As this model emphasizes individual differences as determinants of consumer behavior, it may complement the TPB by adding depth regarding psychological characteristics influencing decisions. SOR theory and the MGDB model consumer behavior as a dynamic interaction between external stimuli (like dark patterns), internal cognitive and emotional processes (the organism/person), and observable responses over time. SOR theory emphasizes the internal processing of stimuli, which can be informed by both the Bias Taxonomy and the 3 M model, as it considers how individual characteristics modulate responses to external stimuli. The Bias Taxonomy categorizes cognitive biases based on individual differences and motivations. This framework complements the previous theories—especially the 3 M model and SOR theory—by providing insights into why certain biases may be more pronounced in some individuals, thereby allowing for mechanistic predictions regarding dark patterns

that rely on cognitive biases (e.g., *Social Engineering* dark patterns) and providing a better theoretical basis to understand consumer vulnerabilities. The FBM aligns with the TPB and the COM-B model by emphasizing the role of motivation and ability as behavioral drivers. Like in SOR theory, the FBM incorporates triggers—external stimuli—as factors that directly influence behaviors. The DIP-theory has a specific focus on deceptive techniques and how these can be identified. It incorporates elements of SOR theory, emphasizing how deceptive practices affect consumer perceptions and behaviors.

An important implication of the reviewed theories concerns the distinction between benign nudges and manipulative practices. Rather than relying on surface-level design features, these theories suggest that the distinction hinges on the psychological mechanisms through which behavior change is achieved. Practices that preserve users' perceived control, align with their intentions, and operate through transparent, reflective processes may be considered benign, whereas practices that obstruct goal-consistent action, exploit automatic or affective responses, or disproportionately target vulnerable users raise concerns of manipulation. This mechanism-based perspective offers a theoretically grounded complement to legal and ethical approaches, and points toward measurable criteria for evaluating dark patterns and similar nudging practices that exploit the same psychological mechanisms empirically (see Duane et al., 2025, for a review of different nudging practices).

Table 1 summarizes the theories, models, and frameworks described in this paper with their key concepts, how they relate to dark patterns research, and highlights some of their strengths and limitations. Dark patterns researchers need not apply one of these theories in isolation. A synthesized model might combine the situational framing of COM-B, the trait-moderated biases of the Bias Taxonomy, and the dynamic feedback loops of SOR and MGDB.

14. Future research direction

The most frequently used theoretical foundations based on the psychology of heuristics and biases and the dual-process model do not explain why (all, if any) dark patterns influence choices. More specific theories to understand the different types of dark patterns are needed. Although the discussed theories provide a broad set of variables to derive testable hypotheses, they do not allow one to mechanistically explain why specific dark patterns have specific effects on consumer choices. The question here is, what exactly do specific dark patterns influence so that consumers make different choices? Is it, for example, perceived control, consumer trust, attitudes, or intentions? Or do some dark patterns affect consumer perceptions of the vendor or the digital platform as a whole such that their choices are influenced via this pathway? And do these influences eventually lead to different decisions and choices? More nuanced theories are needed that delve into the mechanisms that explain the why behind specific dark patterns to help better understand what it is exactly that influences consumers, as not all dark patterns leverage heuristics and biases (e.g., some Obstruction, Sneaking, or Forced Action dark patterns don't rely on these).

Consider the case of scarcity-based dark patterns, which are often treated as a unified category. Research, however, shows that the scarcity-based mechanisms differ substantially across types. The scarcity bias can be the result of a quantified limitation—which can be supply-based (e.g., “limited edition, only 5 pieces left”) or demand-based (e.g., “high demand, only 5 pieces left”)—or a time-based limitation (e.g., “only available for the next hour”). Researchers studied the effectiveness of these three sources of scarcity (supply-based, demand-based, time-based) on the basis of three different theoretical mechanisms (Barton et al., 2022). (1) Supply-based scarcity is often explained based on uniqueness theory (Fromkin, 1970; Snyder & Fromkin, 1977), which suggests that people desire to own scarce or rare items (which were produced or distributed with a limited supply), because it sets them apart from other people and makes them appear more exclusive and unique. Thus, scarce products resulting from limited supply have a high desirability because they may convey a higher social status (Lynn, 1992). This theory thus suggests that supply-based scarcity has a motivational mechanisms that influences desire or self-identify (key components in MGDB). (2) Demand-based scarcity signals a great popularity of the product and serves as an indicator that many other people value this product. High demand induces conformity through informational social influence and leads to a bandwagon effect (Bindra et al., 2022), also known as social proof (MacCoun, 2012; Scott & Barden, 2022) or social norms (Melnik et al., 2022). Briefly summarized, people adopt the attitudes and decisions of the majority because they take this information as evidence about the quality or

Table 1. Summary of theories, frameworks, and models applicable to dark patterns scholarship.

Theory, framework, or model	Key concepts	Application to dark patterns research	Strengths	Limitations
Ajzen's Theory of Planned Behavior (TPB)	Intentions, attitudes, subjective norms, perceived behavioral control	Explains how attitudes, social norms, and control influence responses to dark patterns; e.g., social proof affects subjective norms.	Provides a simple structured approach to understanding consumer behavior from values/norms to intentions to actions.	Limited in addressing the nuanced effects of specific dark patterns, e.g., via heuristics and biases; requires additional integration for dynamic scenarios.
Mowen's Meta-Theoretic Model of Motivation and Personality (3 M)	Hierarchical structure of personality traits: elemental, compound, situational, surface traits	Identifies personality traits that influence susceptibility to dark patterns; e.g., impulsiveness and susceptibility to scarcity tactics.	Integrates motivation and personality; hierarchical approach helps identify traits linked to consumer vulnerabilities.	Focused on personality, less detailed on contextual and situational factors in digital environments.
Jacoby's Stimulus-Organism-Response (SOR) Theory	Dynamic interactions between external and internal factors (Stimulus, Organism, and Response)	Explains how dark patterns act as stimuli affecting cognitive/emotional processing, leading to behavioral responses.	Emphasizes internal processes and individual differences; applicable to heuristic-processing and emotional influences of dark patterns; emphasis on long-term behavioral outcomes and feedback loops	Complex; lacks guidance on specific constructs (operationalization) that might matter to consumer decisions but leaves this interpretation to the researcher applying the theory.
Fogg's Behavior Model for Persuasive Technologies (FBM)	Motivation, ability, triggers	Details how dark patterns may manipulate motivation, ability, and use triggers to influence consumer behavior.	Offers actionable insights into designing and deconstructing dark patterns; widely applicable to digital persuasion.	Simplistic; overlooks individual differences and broader contextual influences; lacks guidance on operationalization of motivation, ability, and triggers.
Oreg's and Bayazit's Bias Taxonomy from an Individual Differences Perspective	Verification, simplification, and regulation biases	Highlights why individual differences affect vulnerability to biases exploited by dark patterns like scarcity and Confirmshaming. Allows integration with any other theory mentioned here.	Focus on individual differences provides nuanced understanding of bias susceptibility; mechanistic insights.	Not comprehensive for dark patterns that exploit collective/social dynamics; not applicable as a stand-alone theory for the effects of specific types of dark patterns.
Xiao's and Benbasat's Deceptive Information Practices Theory (DIP)	Concealment, equivocation, falsification; manipulation of information content, presentation, generation	Links dark patterns like interface interference and hidden information to deception mechanisms. Focuses specifically on deception and is thus highly relevant to deceptive designs research (e.g., the detection thereof).	Offers a typology for categorizing dark patterns based on deception strategies; actionable for studying manipulative tactics and identifying those.	Focused on deception; less comprehensive for emotional and cognitive exploitation in dark patterns; no guidance on effects of specific types of dark patterns; does not include relevant individual differences to study consumer susceptibilities to dark patterns.
Xiao's and Nicholson's Multidisciplinary Cognitive Behavioral Framework (MCB)	Antecedents, triggers, impulse buying act, post-purchase outcomes	Explains how dark patterns provoke impulsive decisions (e.g., scarcity, urgency, social proof) and their post-purchase impacts.	Integrates emotional and environmental triggers with post-purchase effects; highlights the consumer's decision process holistically.	Focused on impulsive buying and thus is not relevant to all types of dark patterns; may not generalize well to non-impulsive consumer behaviors influenced by dark patterns; does not include relevant individual differences to study consumer susceptibilities to dark patterns.
Michie's and West's Behavior Change Wheel (BCW) & COM-B	Capability, opportunity, motivation; intervention strategies	Useful for categorizing dark patterns based on how they impact capability, opportunity, or motivation.	Combines theoretical constructs with practical applications; valuable for mitigation strategies against dark patterns.	Simplistic; overlooks individual differences and broader contextual influences; lacks guidance on operationalization of capability, opportunity, or motivation.
Bagozzi's Model of Goal-Directed Behavior (MGDB)	Goals; desires; emotions; includes a wide range of situational, consumer, and vendor characteristics	Comprehensive framework linking consumer behavior to dark patterns' situational, cognitive, motivational, and emotional triggers; allows a more nuanced view at dark patterns on the low, meso, and high level.	Holistic approach integrating cognitive, emotional, and situational factors; broad inclusion of individual differences and contextual differences; applicable to all types of dark patterns; supports theory-building for specific dark patterns. Builds on TPB at its core.	Complex and resource-intensive for empirical application; broad focus may dilute applicability to specific dark pattern types.

value of the product. This theoretical lens suggests that demand-based scarcity influences perceived norms (a key component in the TPB and MGDB) and perceived value. The mechanisms that explain supply-based and demand-based scarcity, therefore, are different and seem to be independent from each other—yet, both scarcity effects are triggered based on the product's limited quantity. (3) In the case of time-based scarcity, a product is scarce because it is tied to a certain period. A time restriction does not necessarily result from a limited quantity; more often, it is the offer that is scarce, not the product (Aggarwal et al., 2011). Limited time offers create a sense of urgency (Barton et al., 2022), which may lead to more impulsive buying decisions (Abbott et al., 2023; Wu et al., 2021). Impulsive buying (a key component of the MCB framework) may result from people's fear of missing out on an offer or because they may regret later not having made the purchase when the product was available (e.g., Naeem, 2020; Xiao & Nicholson, 2013). Time-based scarcity, hence, exploits yet another psychological mechanism based on emotional responses that appears unrelated to the two mechanisms that supply-based and demand-based scarcity exploit. Each of these scarcity types activates distinct psychological mechanisms, and neither the dual-process model nor heuristics and biases explain or predict why people may react differently to different scarcity messages or why scarcity messages may affect consumer choices at all. It is not clear, nor do the three theories applied to scarcity even mention, whether scarcity messages (de)activate System 1 or System 2 thinking. And the three theories that explain scarcity effects likely do not explain why consumer choices are influenced by dark patterns that are based on visual interface designs, emotional language, pre-selected (default) choices, or other identified tactics to influence consumer choices. In summary, the mechanisms behind why different dark patterns influence choices are not well understood and more nuanced theoretical research on dark patterns is needed (Chang et al., 2024), which should be combined with measures of the factors (variables) that they ostensibly influence. If a dark pattern (e.g., Confirmshaming) is supposed to influence a specific emotional response, it would be useful to measure that emotional response to understand whether this is indeed affected and acts as a mediator between the dark pattern and the consumer's choice. Consequently, we need testable theories tied to specific dark pattern types. This entails:

- Identifying mediators of effects: For example, does Confirmshaming affect guilt, or does Forced Continuity reduce perceived control?
- Mapping design features to mechanisms: e.g., which cognitive, emotional, or social pathways are targeted by each pattern?
- Incorporating moderators: e.g., who is most affected based on individual differences such as impulsivity, self-efficacy, or other traits that may expose susceptibility to specific dark patterns (see Schilling & Sergeeva, 2025 for one attempt of such analyses)?
- Integrating feedback loops and post-choice effects: e.g., how do dark patterns affect regret, satisfaction, brand trust, or future resistance and how does this influence affect future interactions with digital platforms and consumers' choices on these?
- Testing multiple mechanisms side by side: e.g., does a preselected default operate via omission bias, inertia, or social expectation (which are the mechanisms discussed in a meta-analysis on default effects, see Jachimowicz et al., 2019)?

At the institutional level, it is important to note that mechanistic insights into consumer behavior do not—and should not—translate directly into prescriptive interface design mandates. Legal scholarship has emphasized that *ex ante* design regulation is often predictably over- and under-inclusive, and that applying existing consumer protection law *ex post* may, in many cases, be a more appropriate response to deceptive digital practices (Dickinson, 2023; 2024). Accordingly, the contribution of theory in dark patterns research lies not in prescribing regulatory solutions, but in improving analytic precision. By identifying the psychological determinants targeted by persuasive interfaces, theory can support research and detection efforts that distinguish benign nudges from manipulative practices without relying on rigid design rules.

15. Conclusion

By summarizing the theories, frameworks, and models in this article and relating them to dark patterns, I aimed to provide researchers with a sound theoretical foundation to develop empirically testable

hypotheses, refine taxonomies and ontologies of dark patterns, and design theory-guided interventions to study their manipulative effects. Each theory contributes unique insights—ranging from motivation and emotion to social norms and personality traits—thereby enriching the broader discourse on consumer vulnerabilities and ethical design in digital environments. However, these are general theories of consumer behavior. As such, they offer little psychological specificity regarding why people respond to particular dark patterns in the way they do. The field still lacks nuanced, pattern-specific theories that can explain, for instance, what exactly a dark pattern alters. Is it guilt, self-image, perceived control, or something else? A more mechanistic, fine-grained understanding is needed—grounded in theory and backed by data.

The theories reviewed in this article offer a rich menu of testable constructs: desires, attitudes, emotions, identity, intention types (I- vs. We-intentions), perceived control, social norms, heuristics, capability, opportunity, and more. Many of these are already measurable via validated scales or behavioral proxies. Future dark patterns research should harness these constructs to generate mechanistic, falsifiable hypotheses. In doing so, scholars can improve the explanatory precision of their models and help transition the field from a descriptive catalog of interface tricks to a predictive science of digital influence. This shift will ultimately support more targeted detection tools, stronger consumer protection policies, and clearer ethical boundaries in the design of digital environments.

My hope is that this article not only clarified what established theories offer to dark patterns scholars but also offered an impetus for developing or refining new theories—ones tailored to explain the psychological mechanisms of specific dark patterns. Understanding how and why particular designs influence users is not just an academic exercise; it is essential for creating fairer, more respectful digital platforms.

Notes

1. For heuristics, see: [https://en.wikipedia.org/wiki/Heuristic_\(psychology\)](https://en.wikipedia.org/wiki/Heuristic_(psychology)); for a list of cognitive biases, see: https://en.wikipedia.org/wiki/List_of_cognitive_biases.
2. Since experiments typically aim to identify mechanisms and explain why and how variables are causally linked, these studies are particularly relevant for evaluating theory use. The 23 studies reviewed here represent all experimental dark patterns studies I identified through iterative literature searches conducted between January 2024 and September 2025 that employed at least one dark pattern to examine its effects on consumers.
3. See, for example, the journals *Psychology & Marketing*, *Journal of Consumer Psychology*, *Journal of Consumer Research*, *Journal of Consumer Psychology*, *Journal of Marketing Research*, *Journal of Retailing*, *Journal of Consumer Behaviour*. There even exists a journal that is wholly dedicated to Marketing Theory.
4. Ideally, a new consumer theory should build upon a previously published consumer theory, so that a discussion of the theories in historical order makes sense. It often is the case that one theory builds upon another previously published theory, but not always.
5. This model is supposed to be ever evolving, so that the author created a website where he updates the FBM and sells his services related to this model on www.BehaviorModel.org.

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Ethics declarations

No ethics approval was required as this research did not involve data collection.

Author contributions

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ORCID

Thomas (Tomek) Schilling  <http://orcid.org/0000-0002-5150-612X>

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About the author

Thomas Schilling specializes in experimental psychology/economics. His research examines consumer decision-making and consumer psychology. Specifically, he investigates how situational factors (e.g., dark patterns) and stable dispositions (personality traits and other individual differences) interact and jointly shape consumers' choices and behavior. He's happy to collaborate with anyone interested in similar research.